

JOSHUA D. NGUYEN

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PROFESSIONAL SUMMARY

Gameplay engineer with 5+ years of professional experience shipping games and interactive systems in Unreal Engine 4/5 and Unity3D. Hands-on background in C++ gameplay systems, multiplayer replication, NPC/AI behavior, and AAA-level performance optimization — including direct benchmarking work with Riot Games on Valorant. Comfortable prototyping, iterating quickly, and incorporating critical feedback in close collaboration with art and design. Genuinely passionate about games and able to speak critically about what makes existing titles work or fall short.

TECHNICAL SKILLS

Gameplay Systems: Player movement & character behaviors, combat mechanics, Gameplay Ability System (GAS), animation state machines, NPC/AI behaviors, object pooling, component architecture, live service feature development

Multiplayer & Networking: Unreal Engine actor/property replication, RPCs, client/server architecture, online subsystems, Steam Web API, PlayFab, Photon Fusion, AWS/Azure backend, latency optimization

Performance & Optimization: GPU/CPU frame timing analysis, RenderDoc, NDK-level profiling, Android GPU optimization, AAA title benchmarking (Valorant/Riot Games, Cyberpunk 2077, BG3, Forza), rendering pipeline validation, bug fixing & system stability

Languages: C++ (C++17), C#, TypeScript, Python, JavaScript, Blueprint Visual Scripting

Engines & Tools: Unreal Engine 4/5, Unity3D, Perforce, Git, Jira, Azure DevOps, RenderDoc, Visual Studio, GitHub Copilot

Platforms: PC (Steam), Oculus/Meta Quest 2 (Android XR), AWS, Azure

PROFESSIONAL EXPERIENCE

Meta / Facebook | *Gameplay Software Engineer II (Contract)* Sept 2025 – Present

- Developed NPC tooling and dynamic branching dialogue systems in C++ for Horizon Worlds, prototyping and iterating on gameplay behaviors in close collaboration with design and art teams within a live-service engine used by millions of creators.
- Led migration of gameplay and editor components from C# (HUR) to C++ (HSR), maintaining behavioral contracts and system stability across a production engine — satisfying both surface-level feature requirements and deep engine-level correctness.
- Worked on NavMesh systems and NPC callbacks during the runtime migration, fixing bugs in existing pathfinding and lifecycle systems to ensure correct behavior across the C#-to-C++ boundary.
- Refactored component pooling for robustness and performance, increasing code coverage to 75%; iterated based on critical feedback from cross-functional engineering, design, and art reviews.
- Wrote robust, maintainable C++ and TypeScript creator samples and AI training data, demonstrating NPC functionality at simple through advanced complexity levels.

HP Inc. | *Gaming Software AI Engineer* May 2024 – April 2025

- Partnered with Riot Games to benchmark and optimize Valorant performance on HP OMEN hardware, analyzing CPU/GPU frame timing, thermal behavior, and driver-level bottlenecks — direct AAA technical collaboration with a top-tier live-service studio.
- Benchmarked AAA titles including Cyberpunk 2077, Baldur's Gate 3, and Forza Motorsport using automation tooling, developing a detailed, comparative understanding of what separates polished gameplay experiences from rough ones at a design and systems level.
- Prototyped and iterated on gameplay features and AI-driven systems in Unreal Engine 5 (C++17) and Unity3D (C#), using LLM-assisted development to accelerate delivery and explore unconventional solutions.
- Profiled and fixed performance bugs on Android-based VR hardware using NDK-level GPU tooling, addressing native C++ bottlenecks and improving frame pacing across platforms.

- Used RenderDoc for frame-by-frame rendering analysis, validating shader outputs and pipeline behavior to ensure consistent quality across platforms.

Janus Research Group | *Game Content Developer & Technical Level Designer (Mid-Level)*
Sept 2023 – May 2024

- Implemented AI-driven NPC systems for military training simulations in Unity and Unreal Engine — behavior trees, state machines, and pathfinding — collaborating directly with designers and artists, iterating based on feedback to meet functional and experiential goals.
- Iterated on gameplay features and implementations for usability, performance, and bug fixes; built Object Pooling and Screenshot plugin tools to improve creator workflows.
- Developed animation state machines for FBX imports, enabling artists to integrate character motion independently; analyzed crash reports and logs to fix system-level issues.
- Collaborated across programming, art, and design in an Agile/Scrum environment on tight delivery schedules, working independently while staying aligned with team direction.

Combat Waffle Studios | *Full Stack Engineer* April 2023 – Sept 2023

- Shipped Ghosts of Tabor to Steam and Oculus Quest 2 — a 3D VR PVP survival multiplayer game. Implemented Steam Web API, multiplayer server infrastructure, and online subsystem in Unreal Engine with AWS/Azure and Azure PlayFab — direct experience writing multiplayer gameplay systems for a live, shipped title.
- Worked across engine scripting, backend architecture, and cloud infrastructure as part of a small, self-directed team delivering a commercially released title under real production constraints.

Infovision | *AR/VR Engineer (Mid-Level – Contract)* May 2022 – April 2023

- Built multiplayer XR experiences in Unity3D using Photon Fusion, Vivox, and WebGL for enterprise clients including Verizon, AT&T, and T-Mobile; applied NavMesh and Unreal Navigation Mesh for responsive AI behavior.
- Used 3D math (dot product, cross vectors) and .NET/C# for physics-based distance interaction systems; packaged AR/VR applications for Oculus Quest 2 across platforms with native C++ plugin optimization.

Mercer Project & Drastic Studio | *Unreal Engine 4 & Unity3D Engineer (Contract)* Dec 2021 – April 2022

- Built multiplayer gameplay and networking systems in UE4 (C++/Blueprints) and Unity3D (C#), including actor replication, RPCs, and VR multiplayer on Oculus Quest 2; debugged and fixed network systems for latency and bandwidth issues.

Onessus Blockchain Systems | *Unreal Engine 4 Engineer (Contract)* June 2021 – Nov 2021

- Implemented Gameplay Ability System (GAS) for an ARPG in UE4/UE5 with C++ and Blueprints — player abilities, combat mechanics, power interactions, and multiplayer replication for HodiGod pre-alpha.
- Designed and maintained efficient, reusable C++ systems across gameplay, networking, and in-game economy layers, with a focus on long-term maintainability.

Software Decisions Inc. | *Jr. Software Developer* April 2021 – Nov 2021

- Built C# automation tools and SQL Server workflows for financial and healthcare data pipelines; authored stored procedures and SDLC documentation.

Cornerstone Home Lending | *DevOps Support Analyst / Jr. Software Developer* Feb 2020 – April 2021

- Managed CI/CD pipelines in Azure DevOps; wrote C#/C++ utilities for data sync and record management; supported developer and operations tooling across Dev/Test/Production.

EDUCATION

CG Spectrum — BS in Game Development	2025
University of Houston Victoria — BS in Computer Science	2019

NOTABLE PROJECTS

Ghosts of Tabor — Shipped VR PVP survival multiplayer title on Steam and Oculus Quest 2 (Combat Waffle Studios, 2023).

Cyberwar — Personal Unreal Engine 4/5 action game; custom C++ gameplay systems and combat architecture.